

Graph Recursive VCF (grVCF)

A graph-aware VCF for lossless assembly information and unambiguous cohort variants

LinGraph is a variants-guided **pangenome graph builder**. It records variant sequences and assembly coordinates in grVCF and can export cohort calls as a pangenome graph. grVCF uses VCF-like rows plus graph-aware fields and parent-child relationships.

Two representations of a merged cohort share one goal: retain enough sequence, mapping, and variant information to reconstruct the represented assembly regions. The recursive (exact) representation additionally makes every carrier of one row use the same sequence and the same reference location.

Choose the representation

Mode	Where differences live	Same-row guarantee
CIGAR Default merged output	A row groups a representative allele with each carrier's own breakpoint, size, and alignment in FORMAT; TEMPLATEOFFSET can differ.	No. A shared row can cover distinct member sequences and locations.
Recursive / exact --exact	Carriers use the row's allele and placement. Any residual differences are separate child, flank, or SNP rows; children can nest again.	Yes, for the row operation. A complete haplotype allele can still differ after its carried children are applied.

How the hierarchy encodes graph topology

```
reference left flank --> Insertion A --> reference right flank
                        |
                        +--> Child B: nested variant on Insertion A
                            +--> another nested change, if needed
```

A child row's CHROM names its parent allele or shared template, so its coordinates live on that sequence. This allows variants on insertions, deletions, and duplication templates to become explicit graph branches. A graph is a downstream export of the complete grVCF, not its defining purpose.

Scope of losslessness. The representation preserves mapped assembly sequence when all variant classes, nested dependencies, sequence sources, and placement metadata are retained. Unmapped or filtered regions require separate accounting; an SV-only subset or standard VCF projection cannot reproduce all represented bases.

01 Read a grVCF record

grVCF currently uses a VCF 4.2-style envelope and a .vcf extension. INFO describes a representative allele; FORMAT describes each haplotype observation and its assembly location. Read the actual FORMAT column, because caller options and merge stage can change which optional fields appear.

Column or field	Meaning
CHROM / POS / ID	A reference or alternative locus, or a parent variant ID. On a nested row POS is parent-relative. ID may be the sequence target for descendants.
REF / ALT	SNPs use bases; structural rows can use symbolic <INS>, , or <SUB>. INFO/SEQ carries representative inserted or replacement bases.
INFO: SVTYPE, END, SVLEN	Event type, endpoint, and representative signed length. NSUP counts supporting haplotype columns, not sequencing reads.
INFO: SEQ, EXTENDGRAPHCIGAR	Representative sequence and graph alignment. SEQ may be plain bases or a named graph encoding; keep referenced targets.
FORMAT: GT	Haploid state: 1 supports the row, 0 is callable reference, . is missing. Do not read . as 0.
FORMAT: TYPE, SIZE, EXTENDGRAPHCIGAR	Per-observation event, size, and alignment to the representative. CIGAR mode uses these to retain member differences.
FORMAT: ASSEMBLYCONTIG, QUERYCOORD	Source assembly contig and zero-based query coordinate or range, with strand.
FORMAT: TEMPLATEOFFSET	Per-observation breakpoint displacement from row POS. In exact shared placement it is 0; in CIGAR mode it may differ.
FORMAT: BASE, ALLELENAME, LABEL_H	SNP alternate base and optional source/provenance fields. Multiple observations appear as corresponding comma-separated values.

Coordinates and names

SNP POS is one-based. For symbolic root events, the graph converter interprets replacement intervals from POS and END; a pure insertion has END=POS. A nested insertion starts at parent offset POS-1. A nested deletion removes from that offset for abs(SVLEN) bases. Interpret coordinates in the parent named by CHROM, not automatically on the reference chromosome.

New merged IDs use I_, D_, and S_ for insertions, deletions, and SNPs; older files may use INS_, DEL_, and SNP_. A nested insertion can be named I_<parent>_<pos>_<size>, with letter suffixes for distinct sequences at the same position and size. Preserve IDs when copying dependent rows.

02 CIGAR representation: lossless member information

The default cohort merge writes `##graphvcfmergeVersion=cigar`. It chooses one representative allele for a row. Each carrier retains its own size, breakpoint, assembly coordinate, and alignment to that representative. This is a compact, lossless way to describe members without requiring one exact allele per row.

Carrier	Row shown at chr1:100	Own observation saved in FORMAT
h1	Representative insertion A = ACGTTGCA	8 bases at chr1:100; TEMPLATEOFFSET=0; assembly query range identifies the bases.
h2	Same representative row A	10 bases at chr1:105; TEMPLATEOFFSET=5; extended graph CIGAR describes GG after the fourth base.

Illustrative observations above are not literal VCF records. h1 and h2 support one displayed row but differ in both placement and sequence. The FORMAT arrays retain which observation belongs to which haplotype; do not replace them with the representative sequence alone.

What must be kept

- The representative SEQ or its named sequence source, all per-observation EXTENDGRAPHCIGAR fields, TYPE, SIZE, TEMPLATEOFFSET, ASSEMBLYCONTIG, QUERYCOORD, and matching comma-list entries.
- SNP and small-indel files, nested variants and insertion SNPs, plus the reference and local sequence catalogs needed by encoded alleles.
- The caller's `##pseudoLinearMapping` and `##referenceCoverage` lines. Merged output stores the per-sample lines in `cohort.samples.headers.gz` beside the cohort VCFs.

Practical effect

CIGAR mode reconstructs an individual observation from its row plus the observation alignment and placement metadata. The row itself is **not** a universal allele at one breakpoint. Use the recursive / exact merge if downstream graph interpretation needs that stronger row-level guarantee.

Lossless sequence reconstruction and same-row identity are separate properties. CIGAR mode targets the first; recursive / exact mode targets both.

03 Recursive representation: one row, one operation

An exact merge writes `##graphvcfmergeVersion=exact`. It realigns members to a representative placement when possible. A shared row then denotes the same sequence change at the same parent-relative position for every carrier. Differences that remain become separate nested, flank (`_F`), separated (`_S`), or SNP rows.

The next abbreviated example omits FORMAT details. Row A is an insertion at chr1:100 with representative sequence ACGTTGCA. Child B is a **nested variant on Insertion A**; its CHROM is A, and it inserts GG at parent offset 4 (POS=5). S and D are other changes on A.

CHROM	POS	ID	ALT / INFO	Meaning
chr1	100	A	<INS>; SEQ=ACGTTGCA	Parent insertion, 8 bases.
A	5	B	<INS>; SEQ=GG	Child insertion after base 4 of A.
A	3	S	G to A	Child SNP at base 3 of A.
A	3	D	; SVLEN=-2	Child deletion of bases 3-4 of A.

Haplotype	Carried rows	Complete inserted sequence
h1	A	ACGTTGCA
h2	A + B	ACGTGGTGCA
h3	A + S	ACATTGCA
h4	A + D	ACTGCA

All carriers of A agree on A's eight-base sequence and chr1 placement. Their *complete* inserted sequences differ because they carry different child rows. If several haplotypes carry B, B itself has one GG insertion at the same offset on A. A row with CHROM=B can represent a further variant on that inserted GG sequence.

Reconstruction rule

Start from the reference placement, apply the carried top-level row, then apply carried children on the unchanged parent coordinate axis. Recurse into any child allele before placing it into its parent. Sibling coordinates should not be shifted by edits already applied to the string. Strand and QUERYCOORD map the result back to the assembly.

04 What the exact merge changes

The merge reads indexed assemblies and reference FASTAs with `--exact`. It compares shifted observations against the chosen row placement; compatible alleles are realigned there and their remaining bases are represented by child operations. If a move cannot be represented exactly, the member is kept on a separate row.

Situation	Representation
Different bases within a shared insertion	Nested INS/DEL/SUB or insertion SNP rows on the insertion ID; deeper changes recurse on child IDs.
Member deletes fewer bases than the shared deletion	Bases retained by that member are expressed as child insertions on the deletion's coordinate space.
Shifted member, or difference at a row flank	Realign when exact; otherwise use a separate row or <code>_F/_S</code> pieces so one row does not silently mix distinct placements.
Full-locus duplicated sequence	A shared DUP_ template can carry nested variants while preserving copy-specific provenance.

Validation has two independent questions

Question	Current checker output
Can the represented mapped assembly regions be reconstructed?	<code>tools/check_merge_lossless.py</code> reports <code>lossless=yes</code> when its region reconstruction checks pass.
Does each carrier of each row agree with that row's allele and breakpoint?	The same checker reports <code>unified=yes</code> ; mismatches are written to <code>line_check.tsv</code> , including nonzero <code>TEMPLATEOFFSET</code> .

A passing check applies to the files, assemblies, reference catalogs, and mapped regions supplied. It does not assert that unmapped bases or filtered-away variants are represented. The per-sample checker supports plain SEQ; retain encoded sequence catalogs when using other sequence modes.

Sequence encoding versus variant hierarchy

INFO/SEQ may reuse a named target through graph-CIGAR pieces. That is a way to store the *bases* compactly; it does not move the row or replace its child records. `FORMAT/EXTENDGRAPHCIGAR` describes an observation relative to a representative. The two encodings have different roles even when both contain CIGAR-like operations.

05 Produce, inspect, and convert

Run these examples from the repository root. `vcfs.list` has one individual grVCF path per line. `query_paths.txt` has NAME FASTA [FAI] per line, with names matching haplotype columns.

Merge individual grVCFs

```
# CIGAR mode (default)
python tools/merge_grvcfs.py -I vcfs.list -O merged_cigar -t 16

# Recursive / exact mode: indexed assemblies and reference are required
python tools/merge_grvcfs.py -I vcfs.list -O merged_exact -t 16 \
  --exact query_paths.txt --reference-fasta reference.fa
```

The cohort merge writes `cohort.sv.vcf`, `cohort.indel.vcf`, `cohort.snp.vcf`, and `cohort.samples.headers.gz`. Read `##graphvcfmergeVersion=cigar|exact` from each merged VCF. A direct LinGraph graph run also uses `--exact` for its default merged cohort calls.

Check losslessness and shared-row identity

```
python tools/check_merge_lossless.py \
  -v merged_exact/cohort.sv.vcf merged_exact/cohort.indel.vcf \
  merged_exact/cohort.snp.vcf \
  -s samples/*.vcf -r reference.fa -q query_paths.txt -o checks
```

Inspect `checks/summary.tsv` for lossless and unified. The sample VCFs supply their mapping lines; the query FASTAs supply the expected assembly bases. Add `-r` for each local reference template FASTA used by the merge.

Split or change representations

```
python tools/convert_merged_grvcf.py split \
  -v merged_exact/cohort.sv.vcf merged_exact/cohort.indel.vcf \
  merged_exact/cohort.snp.vcf -r reference.fa -o individual

python tools/convert_merged_grvcf.py to-exact \
  -v merged_cigar/cohort.sv.vcf merged_cigar/cohort.indel.vcf \
  merged_cigar/cohort.snp.vcf -r reference.fa -q query_paths.txt \
  --reference-fasta reference.fa -O merged_exact
```

The converter also provides `to-cigar`. Keep `cohort.samples.headers.gz` next to merged files (or pass `--sample-headers`) so split VCFs regain `##pseudoLinearMapping` and `##referenceCoverage`. Equivalent placements in repetitive sequence may yield different row positions on a round trip even when reconstructed alleles agree.

06 Use the complete grVCF

Build a pangenome graph

Feed the complete merged SV, indel, and SNP grVCFs to the graph exporter, with the reference and required sequence sources. Parent and child alleles become graph paths and branches. The rGFA exporter includes all variant sizes by default; a size-filtered or SV-only graph omits parts of the assembly representation.

```
python scripts/merged_vcf_to_gfa.py \  
-v merged_exact/cohort.sv.vcf merged_exact/cohort.indel.vcf \  
merged_exact/cohort.snp.vcf \  
-q query_paths.txt --graph-folder graph -o cohort.gfa -t 8
```

Use the actual graph's local reference templates and alternative sequence catalog when required by your cohort. For LinGraph's full cohort workflow, `--make-graph` exports `cohort.gfa` and per-sample GAFs; `--gfa-only` exports from existing merged calls.

Export a conventional VCF

```
python tools/grvcf_to_vcf.py \  
-i merged_exact/cohort.sv.vcf -o cohort.sv.standard.vcf
```

This is a **representative-allele projection**. The current streaming exporter removes nested rows and POS=0 records, keeps GT and a small INFO subset, and converts an <INS> allele with plain INFO/SEQ bases to explicit REF/ALT. It does not reconstruct each carrier's complete allele, normalize against a reference, or emit an unplaced sidecar. Graph-encoded insertion SEQ stays symbolic. Keep the original grVCFs for lossless work and graph building.

Files and implementation to consult

Purpose	Repository source
Caller schema and per-sample sequence encoding	scripts/graphreftovcf.py; tools/check_vcf_lossless.py
Cohort CIGAR / exact layout, nested rows	scripts/graphvcfmerge.py; scripts/graphvcfmerge_snp_compact.py
Merge and convert cohorts	tools/merge_grvcfs.py; tools/convert_merged_grvcf.py
Reconstruction, graph, standard export	tools/check_merge_lossless.py; scripts/merged_vcf_to_gfa.py; tools/grvcf_to_vcf.py

These are LinGraph grVCF conventions, not an independent VCF standard. Read the file's own header as its field schema and retain the producer version and reference identity when sharing data.